



The Best Formula 1 Driver in the Past Five Seasons

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1: Abstract

Our research explores the question: **Who is the best Formula 1 driver in the past five seasons based on their performance and expertise?** To answer this, we applied a series of statistical modeling techniques, including linear and logistic regressions, fixed effects models, marginal effects analysis, F-tests, and interpretations of Adjusted R-squared and RMSE. Our key variables included driver experience, age, average speed, starting grid positions and more to be explored in our methodology section. Our goal was to isolate individual driver's contributions and performance from external factors like team quality, race-specific conditions, and season-related factors to identify top-performing drivers using our cultivated sets of data. Additionally, by separating skill from external advantages, we could establish a fairer basis for evaluating driver performance across seasons.

Our regression analysis across numerous models revealed a steady improvement in explanatory power with adjusted R^2 increasing from 0.199 to 0.664. By introducing starting grid position and driver fixed effects, our model showed significant improvement in performance, which highlighted that starting position and omitted driver traits are important. Our results demonstrated that experience and age were initially significant but lost predictive value as more independent variables were added (especially with driver identity). However, starting grid position consistently remained the strongest predictor of race outcomes, which suggests to us that starting grid position may be highly correlated with dependent variables. Additionally, the F-test confirmed there have been improvements since earlier models, which concludes us to reveal Lando Norris as the best driver in Formula 1, followed by George Russell and Charles Leclerc for the 2020-2024 seasons, given in our last model (9). The ranking indicates to us that they outperformed their peers even after adjusting for race context.

2: Introduction



Figure 1. Picture of Formula 1 Race

Formula 1 is often considered the highest level of motorsport in the world with the best automotive engineering and technologies. Before the beginning of a season, 20 top drivers are selected to compete under intense and high-stake conditions. Over the past five seasons from 2020 to 2024, the sport witnessed a series of dominant championships by specific teams, the emergence of talented drivers and closely contested races. As a result, these factors make it challenging to determine who truly stands out as the best driver.

Moreover, traditional rankings are often based on total points or championship titles, these measures fail to fully capture individual contribution of each driver, especially when considering external factors such as team performance and race conditions. Our research aims to address the question: **Who is the best formula 1 driver in the past five seasons based on their level of performance and expertise?**

To evaluate driver performance more comprehensively, we set to evaluate three outcome variables: race position, podium finishes and race completion, alongside explanatory variables such as age, experience, and fastest lap times in our modeling. By incorporating both individual skills and external factors such as team strength, track characteristics and seasonal differences

through fixed effect, we hope to provide a more accurate assessment of driver performance and help to understand what really determines success in Formula 1 Racing.

3: Data

Variable	Description
season	Year of the Grand Prix (e.g., 2020, 2021...)
track	Name of the race venue (e.g., <i>austria</i> , <i>monaco</i> , <i>spa</i>)
position	Final race position. A numeric string (e.g., “1”, “2”, “NA”)
starting_grid	Starting position on the grid (numeric)
driver	Full name of the driver
team	Constructor team (e.g., <i>Mercedes</i> , <i>Red Bull</i>)
turns_of_track	Number of corners on the track (numeric)
fastlap	Binary variable: 1 if driver got fastest lap, 0 otherwise
podium	Binary variable: 1 if finished in top 3, 0 otherwise
DNF	Binary variable: 1 if driver did not finish the race, 0 if finished
experience	Total number of Grand Prix entries by the driver up to that race
age	Age of the driver during the race (based on season - birth year)
birthday	Driver’s date of birth in YYYY-MM-DD format
avg_speed	Average Speed coefficient of the driver during the entire race

NOTE: The source of our data was from [GitHub](#), we combined datasets with the names:

1. Formula1_YEARseason_calendar.csv (i.e. Formula1_2022season_calendar.csv)
2. Formula1_YEARseason_raceResults.csv (i.e. Formula1_2022season_raceResults.csv)
3. Formula1_YEARseason_drivers.csv. (i.e. Formula1_2022season_drivers.csv)

We only combined selected columns from certain data sets, not all columns from all of the files mentioned above were included in the final data set used for analysis.

4: Methodology

To determine the best Formula 1 driver, we constructed a series of statistical analyses, such as logit regressions, linear regressions, fixed effects, and marginal effects analysis, which capture the relationships between the independent variables reflecting both performance outcomes and the contextual factors and the dependent variables. Our dataset explored multiple seasons, from 2020 to 2024, and included information related to driver characteristics to consider external factors that influence driver's performance.

Independent Variables

Variable	Description
Experience	Measures how many Grand Prix a driver has competed in up to the current race. Experience may influence performance through greater race knowledge, strategic awareness, and comfort under pressure. However, its effect may plateau or decline at a certain point, meaning there is a need for a quadratic term.
Age	Reflects both physical condition and cognitive decision-making abilities. Younger drivers may be more aggressive and react sharply, while older drivers may compensate with maturity and strategy. Age alone helps us test if youth is a consistent advantage.
Experience ²	Captures the non-linear relationship between experience and performance. While rookies tend to improve with more races, veterans may eventually face decline due to their age or changing motivation. This squared term helps detect performance peaks in a driver's career journey.
Age ²	Similarly to experience ² , this captures a potential turning point in the relationship as it may indicate a performance curve improving with age to a point, then a decline.
Starting Grid	Starting closer to the front increases the likelihood of finishing at better positions and achieving podiums due to less traffic and more control. This helps separate race skill from qualifying skill.

Average Lap Speed	Indicates in-race pace and consistency, which can reveal raw performance ability across varying conditions. It reflects the driver's efficiency, adaptability, and control over the car during the full race distance.
Fastest Lap	Indicates whether the driver achieved the fastest lap in the race and signals their peak performance, which could be used to measure raw speed potential and competitiveness.
Driver FE (i)	Controls for each driver's inherent ability, such as their reflexes, decision-making, adaptability, and temperament that don't fluctuate across races. These are essential to isolate who the driver is, beyond team support, race-specific conditions or seasons.
Team FE (f)	Controls for the car's performance and team resources, such as R&D, engineering quality, and pit stop strategy. Including this allows us to assess driver impact independent of machinery and engineering, which is critical when comparing drivers across different teams.
Season FE (j)	Captures variability across seasons due to changes in technical regulations, tire compounds, car designs, and team competitiveness. Some seasons heavily favour certain teams or driving styles but this fixed effect neutralizes those contextual swings.
Race FE (t)	Controls for the unique characteristics of each Grand Prix, such as circuit layout, weather, humidity, and number of turns that can influence outcomes. These effects help us focus on driver performance across different types of challenges by controlling for event-specific variance.

* FE = Fixed Effect

Dependent Variables:

Variable	Description
Position	A numeric variable that represents the final finishing position of a driver in a race. For modeling purposes, DNF values are often treated as missing or encoded accordingly. This is the most direct measure of race performance as it reflects how well a driver performed relative to the entire race, consisting of overtaking, race pace, strategy, and consistency. Lower numeric positions suggest stronger performance.

Podium	A binary variable equal to 1 if a driver finished in the top three, 0 if otherwise. Podium finishes are symbolic of high performance and achieving them consistently across varying conditions, tracks, and seasons indicates a driver's ability to compete rigorously and successfully. This is useful for highlighting race-winning or front-running capability, and makes for powerful classification modeling.
Completion	A binary variable equal to 1 if the driver finished the race, 0 if otherwise. Race completion is a crucial metric to assess consistency in driver's performance. However, some DNFs are due to mechanical failures, but often result from driver errors like crashes, penalties, and pushing beyond limits. A completion rate often reflects situational awareness, strengths, and the ability to make effective decisions even in difficult scenarios.

One of the statistical modeling techniques we conducted to evaluate driver's performance in Formula 1 was **linear regression**, specifically to interpret the coefficients, significance, and correlation between each variable and external factor when we assign Position (ranking) as one of the dependent variables. Linear regression provides clear, interpretable coefficients that allow us to evaluate how much a variable like experience or starting position shifts a driver's expected finishing position. A negative coefficient implies better performance (lower race position), while a positive one suggests poorer results. For example, if the coefficient for age is positive and significant, it means older drivers are typically associated with finishing the race slower than younger drivers due to their lower physical resistance towards intense gravity and speed. This linear model forms the backbone of our analysis and helps to quantify the direct impact of measurable race and driver characteristics on race results.

Second statistical model technique we used was **logit models** to predict binary outcomes, such as podium placement and race complementation. For these outcomes, we use logistic regression models because logit models estimate the probability of a binary event occurring, given a set of predictors. Podium as a dependent variable helps us identify which factors

significantly influence a top-3 finish. Variables such as experience, age, and fastest lap speed are included to assess their predictive power. Lastly, race completion as another one of the dependent variables explores what determines whether a driver completes the race (finishes vs. DNF, etc).

Thirdly, we included **fixed effects** variables and analysis to control for omitted bias, as regression models are often prone to omitted variable bias where some unobserved factors influence the dependent variable, leading to misleading coefficient estimates. To ensure that the coefficients are accurately explained and represented without omitted bias, we applied fixed effects models to control for unmeasured but consistent characteristics across four factors: driver, team, season, and race fixed effects. **Driver fixed effects** capture stable, unobservable traits like the driver's natural talent, consistency, and decision-making skills. **Race fixed effects** demonstrate track-specific characteristics, such as circuit layout, weather conditions, safety car frequency, and overtaking difficulty. These factors can drastically alter race outcomes and affect drivers' performance. Season fixed effects account for year-to-year variability, including regulation changes, team development cycles, and evolving competition levels. A driver might benefit from a strong car in one season but struggle in the next season due to technical rule changes. Lastly, **team fixed effects** control for the structural advantages of teams including car performance, budget, pit crew, and strategic expertise. By introducing team fixed effects, we can isolate the driver's contribution to success independently to whether they were in a dominant car or not, which is an essential step in identifying who is the best Formula 1 driver in the given seasons.

Fourthly, we conducted **marginal effect analysis** to better understand how each variable contributes to our model and improves predictive accuracy. Marginal effect analysis was implemented by adding a new variable to the model incrementally. This analysis is essential to

determine who the best driver is because it allows us to test not just whether a variable matters, but how much it matters when other variables are already controlled. It also reveals interactions and shifts in significance as more context is added to the model. This technique also helps identify multicollinearity and suppressor effects, ensuring that our model is both statistically valid and conceptually insightful.

Finally, we assessed model quality and significance using **F-test, adjusted R², and Root Mean Squared Error (RMSE)**. F-test is conducted to compare models with different fixed effects and whether inclusion of fixed effects significantly improves the model fit. This statistical test helps determine whether the more complex model with more fixed effects explains significantly more variance than a simpler model. Additionally, adjusted provides us with a measure of explanatory power while penalizing for model complexity with irrelevant predictors. As we add more variables, especially fixed effects, adjusted R-squared can help ensure that the variables are not overfitting. A higher adjusted R-square will indicate that the additional variables explain more of the outcome's variance in meaningful ways. Lastly, RMSE captures the average prediction error of our model and it is a valuable tool to measure because it allows us to evaluate how well our model performs out-of-sample, making it especially useful when we compare models with different combinations of predictors and fixed effects.

Statistical Models:

A) Grand prix position as a dependent variable

1. $Position_{ijt} = \beta_1(Experience_{ijt}) + \beta_2(Age_{ijt})$
2. $Position_{ijt} = \beta_1(Experience_{ijt}) + \beta_2(Experience_{ijt})^2 + \beta_3(Age_{ijt}) + \beta_4(Age_{ijt})^2$
3. $Position_{ijt} = \beta_1(Experience_{ijt}) + \beta_2(Experience_{ijt})^2 + \beta_3(Age_{ijt}) + \beta_4(Age_{ijt})^2$

$$+ \beta_5(\text{Starting Grid}_{ijtf})$$

$$4. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf})$$

$$5. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf}) + \beta_7(\text{Fast Lap}_{ijtf})$$

$$6. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf}) + \beta_7(\text{Fast Lap}_{ijtf}) \\ + [\text{Driver Fixed Effect}]_i$$

$$7. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf}) + \beta_7(\text{Fast Lap}_{ijtf}) \\ + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j$$

$$8. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf}) + \beta_7(\text{Fast Lap}_{ijtf}) \\ + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j + [\text{Season Fixed Effect}]_t$$

$$9. \text{Position}_{ijtf} = \beta_1(\text{Experience}_{ijtf}) + \beta_2(\text{Experience}_{ijtf})^2 + \beta_3(\text{Age}_{ijtf}) + \beta_4(\text{Age}_{ijtf})^2 \\ + \beta_5(\text{Starting Grid}_{ijtf}) + \beta_6(\text{Average Lap Speed}_{ijtf}) + \beta_7(\text{Fast Lap}_{ijtf}) \\ + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j + [\text{Season Fixed Effect}]_t \\ + [\text{Team Fixed Effect}]_f$$

B) Podium as a dependent variable

* $Podium = \{1 \text{ if podium, } \emptyset \text{ if otherwise}\}$

$$10. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Age_{ijtf})$$

$$11. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2$$

$$12. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf})$$

$$13. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf}) + \beta_6(Average\ Lap\ Speed_{ijtf})$$

$$14. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf}) + \beta_6(Average\ Lap\ Speed_{ijtf}) + \beta_7(Fast\ Lap_{ijtf})$$

$$15. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf}) + \beta_6(Average\ Lap\ Speed_{ijtf}) + \beta_7(Fast\ Lap_{ijtf}) \\ + [Driver\ Fixed\ Effect]_i$$

$$16. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf}) + \beta_6(Average\ Lap\ Speed_{ijtf}) + \beta_7(Fast\ Lap_{ijtf}) \\ + [Driver\ Fixed\ Effect]_i + [Race\ Fixed\ Effect]_j$$

$$17. Podium_{ijtf} = \beta_1(Experience_{ijtf}) + \beta_2(Experience_{ijtf})^2 + \beta_3(Age_{ijtf}) + \beta_4(Age_{ijtf})^2 \\ + \beta_5(Starting\ Grid_{ijtf}) + \beta_6(Average\ Lap\ Speed_{ijtf}) + \beta_7(Fast\ Lap_{ijtf}) \\ + [Driver\ Fixed\ Effect]_i + [Race\ Fixed\ Effect]_j + [Season\ Fixed\ Effect]_t$$

$$\begin{aligned}
18. \text{Podium}_{ijt} \{1 \text{ if podium, } \emptyset \text{ if otherwise}\} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 \\
&+ \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
&+ [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j + [\text{Season Fixed Effect}]_t \\
&+ [\text{Team Fixed Effect}]_f
\end{aligned}$$

C) Completing the race as a dependent variable

* *Completion* = {1 if completed the race, $\emptyset$ if otherwise, such as DNF}

$$\begin{aligned}
19. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Age}_{ijt}) \\
20. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
21. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
&+ \beta_5(\text{Starting Grid}_{ijt}) \\
22. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
&+ \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) \\
23. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
&+ \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
24. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
&+ \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
&+ [\text{Driver Fixed Effect}]_i \\
25. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2
\end{aligned}$$

$$\begin{aligned}
& + \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
& + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j
\end{aligned}$$

$$\begin{aligned}
26. \text{Completion}_{ijt} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 + \beta_3(\text{Age}_{ijt}) + \beta_4(\text{Age}_{ijt})^2 \\
& + \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
& + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j + [\text{Season Fixed Effect}]_t
\end{aligned}$$

$$\begin{aligned}
27. \text{Completion}_{ijt} \{1 \text{ if completed, } \emptyset \text{ if otherwise}\} &= \beta_1(\text{Experience}_{ijt}) + \beta_2(\text{Experience}_{ijt})^2 \\
& + \beta_5(\text{Starting Grid}_{ijt}) + \beta_6(\text{Average Lap Speed}_{ijt}) + \beta_7(\text{Fast Lap}_{ijt}) \\
& + [\text{Driver Fixed Effect}]_i + [\text{Race Fixed Effect}]_j + [\text{Season Fixed Effect}]_t \\
& + [\text{Team Fixed Effect}]_f
\end{aligned}$$

5: Results

To accurately identify the best Formula 1 driver based on both performance and expertise, this study applies a sequence of regression models, each improving on the last by introducing additional explanatory variables and structural enhancements. The regression analysis across Models 1 through 9 demonstrates a clear improvement in explanatory power, with Adjusted R^2 increasing from 0.199 in Model 1 to 0.664 in Model 9. Notably, two critical inflection points emerge: Model 3 shows a substantial gain in fit with the inclusion of starting grid position, while Model 6 significantly enhances explanatory strength through the addition of driver fixed effects. These milestones emphasize the dual importance of both track context and driver-specific ability in shaping race results.

Individual-level predictors evolve throughout the models. Experience initially matters in Models 1-2 but rapidly loses significance, suggesting that once identity is controlled for, tenure alone no longer explains performance. This is reinforced in Models 6-9, where driver fixed effects absorb the variation previously attributed to experience. Similarly, age, initially a strong positive predictor (older drivers performing worse) fluctuates and eventually disappears from the models altogether, likely due to multicollinearity with fixed effects. Neither experience² nor age² offer evidence of nonlinear effects. These results collectively indicate that who the driver is matters more than how old or experienced they are.

Conversely, starting grid position stands out as the most robust and stable predictor. Introduced in Model 3, it maintains a large, significant impact across all subsequent models, including the fully saturated Model 9. Average speed, while initially insignificant, gains predictive power in Models 7-9 once key fixed effects are controlled, implying that faster drivers do finish better after adjusting for team and race characteristics. Meanwhile, fastlap remains

unstable and largely redundant throughout. These insights are further visualized through a series of fixed effects plots under the **6: Visualizations** Section.

Horizontal Bar Plot 1 illustrates Driver Fixed Effects from Model 9 across all races. Drivers like Oliver Bearman, Franco Colapinto, and Liam Lawson appear to outperform expectations, while seasoned veterans like Fernando Alonso, Romain Grosjean, and Lewis Hamilton rank lower, likely due to accumulated disadvantages over multiple seasons or team effects.

To ensure fairer comparisons, ***Horizontal Bar Plot 2*** includes only drivers with full-season participation from 2020 to 2024. This filter removes biases caused by partial-season drivers with small sample sizes. Under this consistent exposure, Lando Norris and George Russell emerge as consistently high performers, while Alonso remains at the bottom, reinforcing the need to evaluate both raw performance and long-term consistency.

Horizontal Bar Plot 3 turns to Team Fixed Effects and shows that Red Bull Racing Honda Rbpt consistently outperforms the field, whereas Alfa Romeo Racing Ferrari underperforms significantly. These results affirm that team machinery and structure play a crucial role and must be accounted for to isolate driver expertise.

To more accurately match fixed-effect results with real-world driver evaluations, a focused analysis on the 2024 season is introduced.

Horizontal Bar Plot 4 presents Driver Fixed Effects for only those who participated in 2024. This approach removes distortions caused by long-term team performance or partial exposure. As expected, rookie outliers from Plot 1 regress toward the mean, and top drivers in the actual 2024 standings like Max Verstappen and Charles Leclerc rise in the fixed effect

rankings. This validates the fixed-effects framework when properly constrained and demonstrates its utility in approximating real-world outcomes.

How does FE stack up against Average Speed?

We evaluate whether fixed effects (e.g., driver, race, team) explain more variance in finishing position than average speed.

Model 4, which adds `avg_speed_coef`, improves model fit moderately (Adjusted $R^2 = 0.56$).

Model 6, with driver fixed effects, achieves a higher Adjusted $R^2 = 0.63$.

In Model 9, `avg_speed` becomes statistically significant only after including full fixed effects, suggesting its earlier insignificance was due to omitted variable bias.

Conclusion: While average speed reflects short-term performance, fixed effects capture persistent traits (driver skill, team quality, track difficulty) and offer stronger explanatory power. However, both are complementary: fixed effects control unobservables, while `avg_speed` provides interpretable insights into race-day performance.

Does Starting Grid effect Driver Performance?

Starting position as dependent variable and exclude starting position in model to see how does out model perform (max always start at the front, hard to improve poll position, so he doesn't have strong effect from starting position)

We test the hypothesis that starting position may not strongly explain Verstappen's performance, as he typically starts in front (P1–P3), and thus has little room for “improvement”.

1. We fit a model with starting_grid as the dependent variable to understand what predicts starting position.
2. We re-estimate our main performance model without starting_grid to see how much predictive power is lost.

```
summary(model_with_grid)$adj.r.squared
## [1] 0.7437939

summary(model_wo_grid)$adj.r.squared
## [1] 0.6722191

sqrt(mean(model_with_grid$residuals^2))
## [1] 1.931556

sqrt(mean(model_wo_grid$residuals^2))
## [1] 2.188717
```

With starting_grid: Adjusted $R^2 = 0.6646$,

RMSE = 2.34

Without starting_grid: Adjusted $R^2 =$

0.6412, RMSE = 2.49

This suggests starting position contributes to explaining race outcomes, but its absence does not collapse model performance, especially for drivers like Max Verstappen who consistently start from the front.

Table 1. Summary Table of Regression Analysis from Model 1 to 9 (Appendix 7)

Model	Fixed Effect	Experience	Expsquare	Age	Age square	Starting Grid	Avg Speed	Fastlap	Adj R
Model 1	-	-0.0697	—	1.1832	—	—	—	—	0.199
Model 2	-	-0.0976	7.51×10^{-5}	3.37	-0.0354	—	—	—	0.205
Model 3	-	-0.0008	-5.40×10^{-5}	-0.512	0.0145	0.668	—	—	0.609
Model 4	-	-0.0187	-1.78×10^{-5}	0.169	0.0048	0.501	1.96×10^{-4}	—	0.572
Model 5	-	-0.0187	-1.78×10^{-5}	0.169	0.0048	0.501	1.96×10^{-4}	-1.231	0.564
Model 6	Driver	-0.024561	-0.000074	-1.346	0.045824	0.347512	-0.00003	-0.2076	0.62901
Model 7	+ Race	-0.015129	-0.000089	Removed	0.04867	0.325769	-0.002383	-0.229	0.651052
Model 8	+ Season	-0.015129	-0.000089	Removed	0.04867	0.325769	-0.002383	-0.229	0.64991
Model 9	+ Team	-0.060666	0.000128	-1.48631	-0.016052	0.290706	-0.00221	-0.0869	0.664578

Table 2. Summary Table of F-Test (Appendix 8)

Model Compared	F-Statistic	p-value	Interpretation
Model 2 vs Model 1	26.94	< 0.001 ***	Experience and age (including nonlinear terms) significantly improve model fit.
Model 3 vs Model 1	87.68	< 0.001 ***	Starting grid position adds substantial explanatory power.
Model 4 vs Model 1	72.95	< 0.001 ***	Average speed significantly contributes to explaining position.
Model 5 vs Model 1	72.95	< 0.001 ***	Fastlap adds minimal new value beyond average speed and starting grid.

Table 3. Summary Table of Adjusted A^2 and RMSE (Appendix 9) for FE Models

Model	Fixed Effects Included	Adj. R²	RMSE	Interpretation
6	Driver	0.629	2.59	Captures talent & style differences across drivers
7	Driver + Race	0.651	2.41	Controls for track-specific features
8	Driver + Race + Season	0.650	2.41	Season adds little new info (possibly redundant)
9	+ Team (Full Model)	0.665	2.34	Best performing model; team matters significantly

Table 4. Top 5 Formula 1 Drivers in 2024 Based on Fixed Effects (Appendix 10)

Top 5 F1 Drivers in 2024 (Based on Fixed Effects)		
Rank	Driver	Fixed Effect
1	 Fernando Alonso	-387.12
2	Lewis Hamilton	-340.56
3	Sergio Perez	-265.11
4	Daniel Ricciardo	-236.79
5	Max Verstappen	-192.50

Table 1 (Appendix 7) summarizes regression progression, while Table 2 (Appendix 8) confirms through F-tests that Models 1-5 significantly improve on the null, with Model 3

offering the largest leap due to grid position. For Models 6-9, high-dimensional fixed effects preclude F-tests; instead, performance is evaluated through Adjusted R^2 and RMSE in **Table 3 (Appendix 9)**. Model 9 consistently performs best, capturing unobserved talent and team context.

The final evaluation compares fixed effect results with actual championship outcomes. While Fernando Alonso appears as the best driver based on his 2024 fixed effect (most negative impact on race position), this does not align with the reality as Max Verstappen was the actual world champion. This discrepancy is instructive meaning fixed effects measure relative driver impact, adjusting for team, track, and grid advantages. In contrast, F1 awards nonlinear points (e.g., 25 for 1st, 18 for 2nd) based on cumulative performance. A driver like Verstappen, who consistently finishes 2nd, may accumulate more points despite slightly weaker fixed effects.

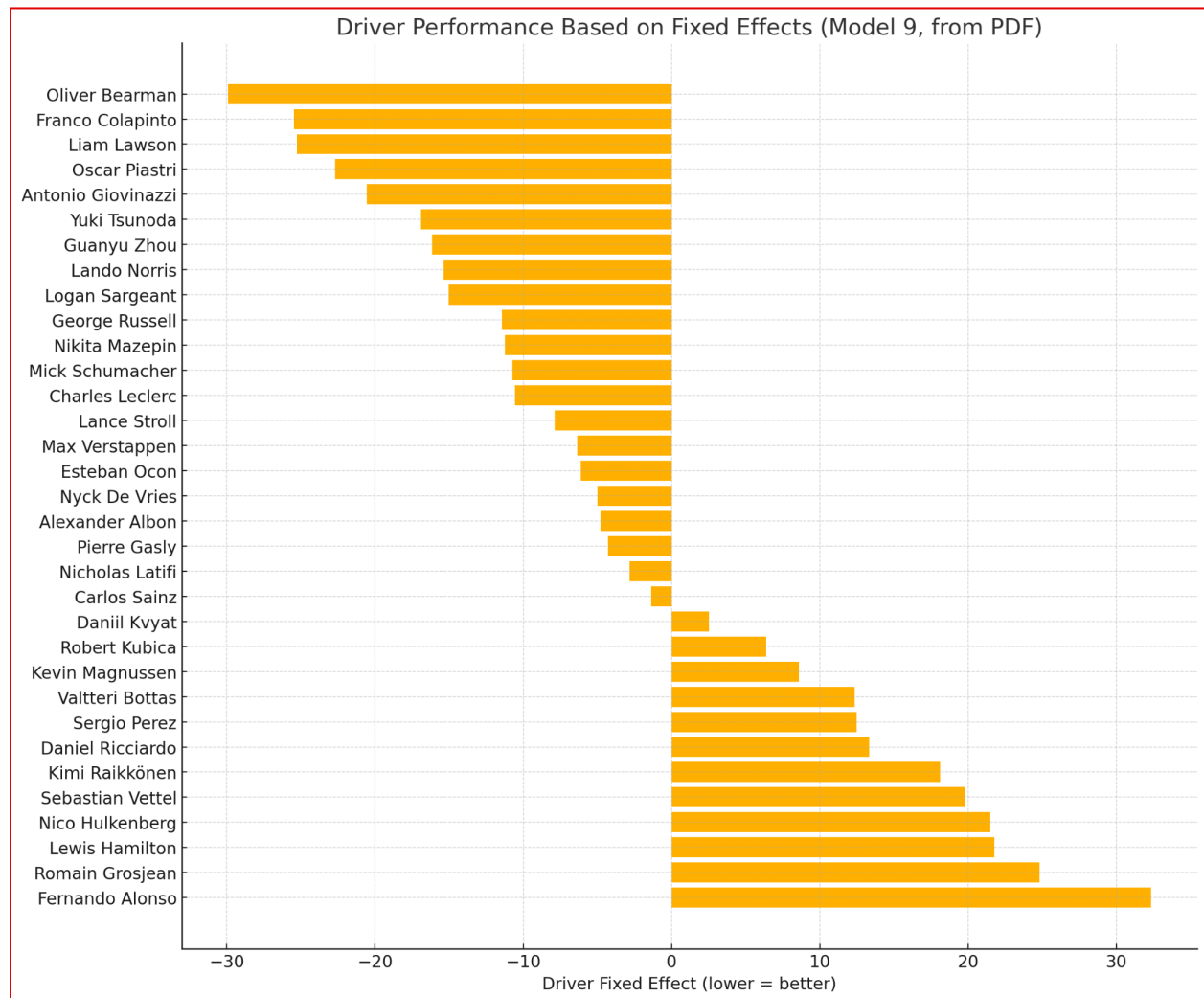
Furthermore, the team advantage enjoyed by Red Bull is absorbed by team fixed effects, diminishing the visible contribution of Verstappen's individual performance. Therefore, while fixed effects provide a deeper view of skill relative to context, they should not be mistaken for championship point proxies.

Table 4 (Appendix 10) summarizes the top 5 drivers in 2024 by fixed effect, offering a focused, season-specific benchmark of driver quality.

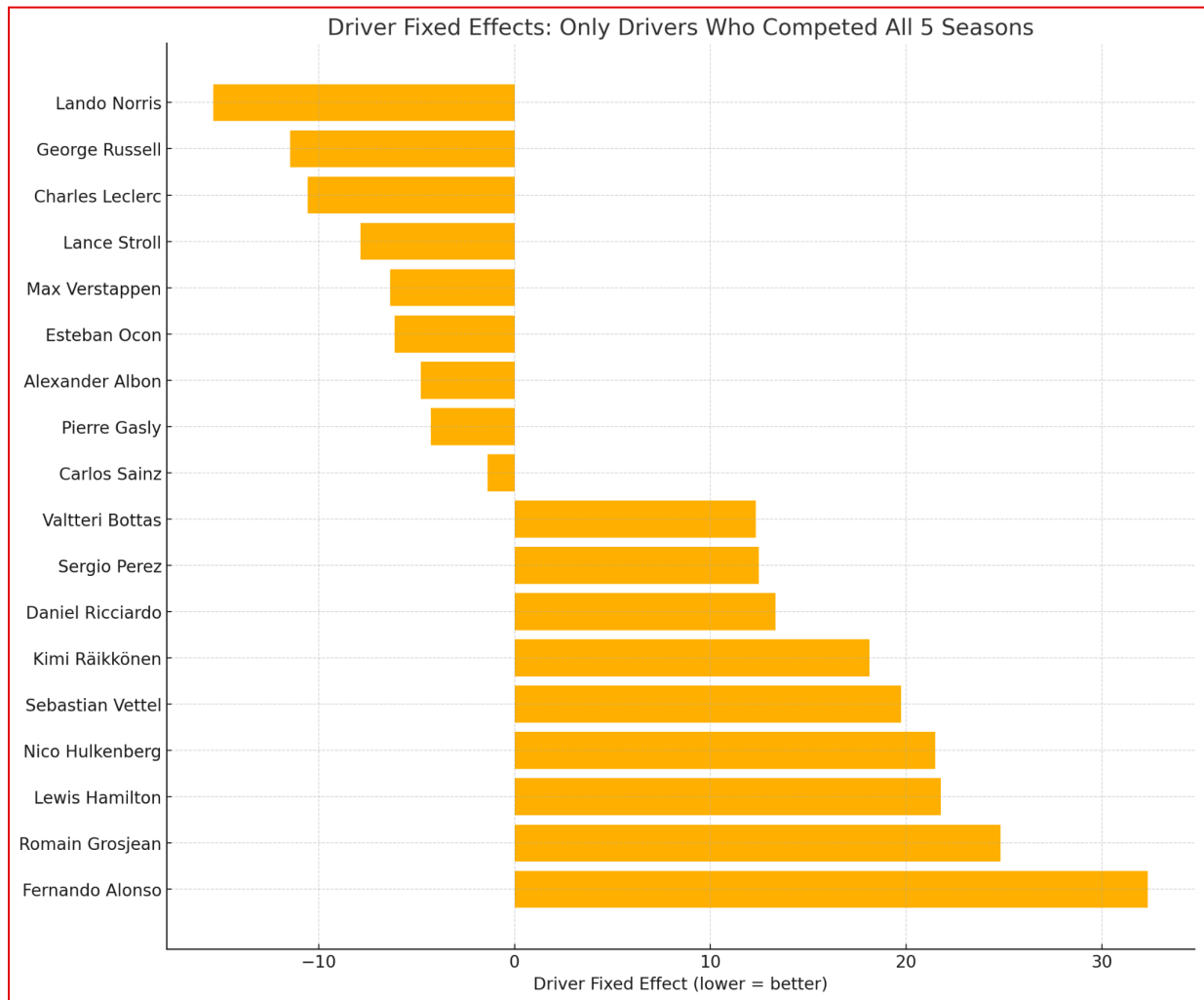
In conclusion, this analytical framework demonstrates that statistical modeling, when properly structured, can reveal performance truths hidden by point tallies. Fixed effects offer a context-aware lens through which driver expertise is evaluated independently of machinery or race conditions. While they do not replace the championship, they allow us to assess who truly drives the performance in Formula 1.

6: Visualizations

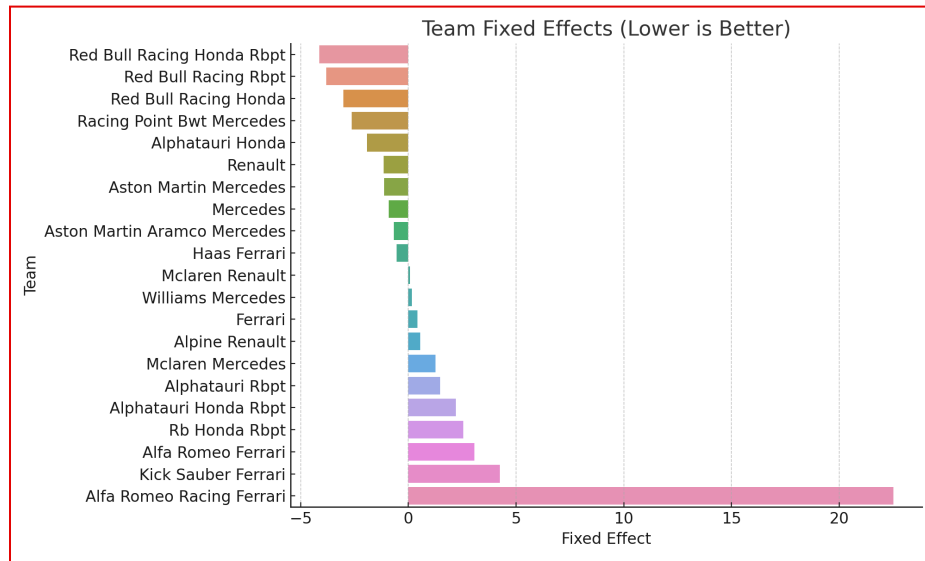
Horizontal Bar Plot 1: Driver Performance Based on Fixed Effect



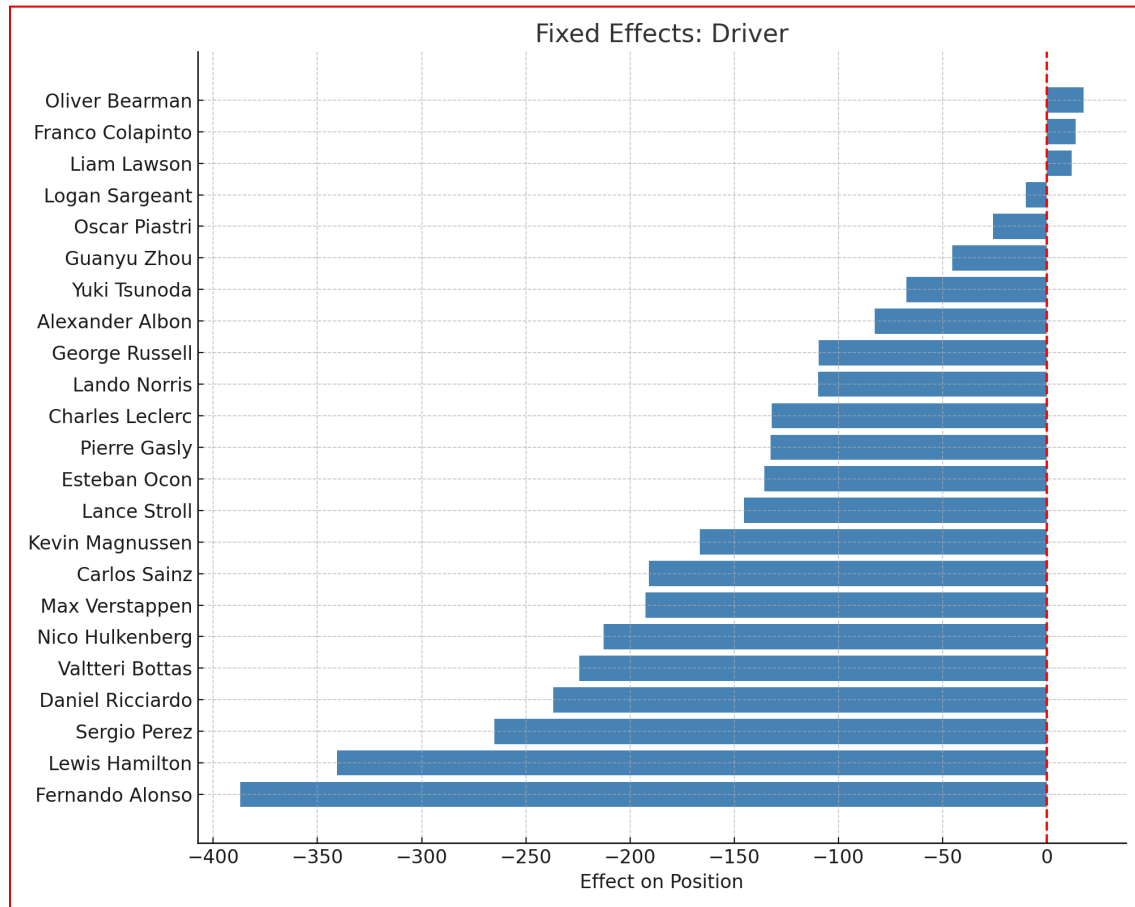
Horizontal Bar Plot 1 shows the visualization of Driver Fixed Effects from Model 9, this plot is sorted from best to worst drivers. This plot will help us determine who is the best driver based on adjusted performance, independent of team, track or season influences.

Horizontal Bar Plot 2: Driver Fixed Effect Based for Drivers with Full Season Participation

Horizontal Bar Plot 2 shows the visualization of Driver Fixed Effects from Model 9 after filtering for drivers with full-season participation for 2020 to 2024. This plot helps ensure a fair comparison based on consistent exposure to race conditions and team changes.

Horizontal Bar Plot 3: Team Fixed Effect with Full Season Participation

Horizontal Bar Plot 3 visualizes Team Fixed Effects from the full-sample regression, reflecting team performance after controlling for driver ability and season-specific factors.

Horizontal Bar Plot 4: Driver Fixed Effect Based for Drivers with 2024 Season Participation

Horizontal Bar Plot 4 visualizes Driver Fixed Effects with 2024 season participation, highlighting individual performance in 2024 season.

7: Summary and Conclusion

While our fixed effects regression model identified Fernando Alonso as the top-performing driver in 2024, this result does not align with the real-world outcome in which Max Verstappen won the World Championship. This discrepancy highlights the nature of our model as the fixed effects capture a driver's average impact on race results after controlling for variables, such as starting grid position, team, track characteristics, and race-specific conditions. As a result, our model may undervalue a driver like Verstappen, who consistently achieves podiums and accumulates championship points but may not stand out as strongly in terms of fixed effects because his Red Bull's dominant car is absorbed through team-specific controls in our analysis.

Additionally, our research treats race position as a linear variable, whereas the official Formula 1 points system is nonlinear and disproportionately rewards race wins. This structural difference further explains why our statistical ranking may differ from the championship outcome. Overall, while our model offers valuable insights into relative driver performance within a controlled context, it is not a substitute for actual race points or standings. Instead, it should be seen as a complementary tool that highlights over or under-performance after accounting for contextual factors.

Our future research can be significantly improved by incorporating a more extensive and detailed dataset. While our current model accounts for several important variables, it lacks specific race-level factors that are crucial in determining driver performance, including the number and timing of pit stops, track characteristics like the number of turns or elevation changes, and weather conditions during each race. By incorporating more detailed variables, it would allow for a more rigorous and statistically accurate analysis and can help isolate a driver's

true contribution from external influences. Additionally, accounting for in-race events such as safety car deployments, mechanical failures, or number of collisions could further refine the model's ability to evaluate performance. By capturing these specific variables, our future models would be better positioned to reflect the complexity of Formula 1 racing and provide a more precise assessment of driver skill and performance ranking.

8: References (APA Citation)

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<https://www.npr.org/2024/11/26/g-s1-35866/formula-1-expands-add-cadillac>.

9: Appendix

Appendix 1. Regression Analysis Code

[MGEC45_GP_v2.html](#)

Appendix 2. Formula 1 Season 2024 Complete Dataset

[F1 Season 2024 Complete Data Set.csv](#)

Appendix 3. Formula 1 Season 2023 Complete Dataset

[F1 Season 2023 Complete Data Set.csv](#)

Appendix 4. Formula 1 Season 2022 Complete Dataset

[F1 Season 2022 Complete Data Set.csv](#)

Appendix 5. Formula 1 Season 2021 Complete Dataset

[F1 Season 2021 Complete Data Set.csv](#)

Appendix 6. Formula 1 Season 2020 Complete Dataset

[F1 Season 2020 Complete Data set.csv](#)

Appendix 7. Summary Table of Regression Analysis from Model 1 to 9 (Table 1)

Model	Fixed Effect	Experience	Expsquare	Age	Age square	Starting Grid	Avg Speed	Fastlap	Adj R
Model 1	-	-0.0697	—	1.1832	—	—	—	—	0.199
Model 2	-	-0.0976	7.51×10^{-5}	3.37	-0.0354	—	—	—	0.205
Model 3	-	-0.0008	-5.40×10^{-5}	-0.512	0.0145	0.668	—	—	0.609
Model 4	-	-0.0187	-1.78×10^{-5}	0.169	0.0048	0.501	1.96×10^{-4}	—	0.572
Model 5	-	-0.0187	-1.78×10^{-5}	0.169	0.0048	0.501	1.96×10^{-4}	-1.231	0.564
Model 6	Driver	-0.024561	-0.000074	-1.346	0.045824	0.347512	-0.00003	-0.2076	0.62901
Model 7	+ Race	-0.015129	-0.000089	Removed	0.04867	0.325769	-0.002383	-0.229	0.651052
Model 8	+ Season	-0.015129	-0.000089	Removed	0.04867	0.325769	-0.002383	-0.229	0.64991
Model 9	+ Team	-0.060666	0.000128	-1.48631	-0.016052	0.290706	-0.00221	-0.0869	0.664578

Appendix 8. Summary Table of F-Test (Table 2)

Model Compared	F-Statistic	p-value	Interpretation
Model 2 vs Model 1	26.94	< 0.001 ***	Experience and age (including nonlinear terms) significantly improve model fit.
Model 3 vs Model 1	87.68	< 0.001 ***	Starting grid position adds substantial explanatory power.
Model 4 vs Model 1	72.95	< 0.001 ***	Average speed significantly contributes to explaining position.
Model 5 vs Model 1	72.95	< 0.001 ***	Fastlap adds minimal new value beyond average speed and starting grid.

Appendix 9. Summary Table of Adjusted A^2 and RMSE (Table 3)

Model	Fixed Effects Included	Adj. R²	RMSE	Interpretation
6	Driver	0.629	2.59	Captures talent & style differences across drivers
7	Driver + Race	0.651	2.41	Controls for track-specific features
8	Driver + Race + Season	0.650	2.41	Season adds little new info (possibly redundant)
9	+ Team (Full Model)	0.665	2.34	Best performing model; team matters significantly

Appendix 10. Top 5 Formula 1 Drivers in 2024 Based on Fixed Effects (Table 4)

Top 5 F1 Drivers in 2024 (Based on Fixed Effects)		
Rank	Driver	Fixed Effect
1	🏆 Fernando Alonso	-387.12
2	Lewis Hamilton	-340.56
3	Sergio Perez	-265.11
4	Daniel Ricciardo	-236.79
5	Max Verstappen	-192.50

10: Figures

Figure 1. Picture of Formula 1 Race (Photo Citation in the Reference Section)

